

AJIO – E-Commerce Order Management Database System

1.Database Design:

Customer Table

Attribute	Description
CUSTOMER_ID	Unique customer ID
FIRST_NAME	Customer's first name
LAST_NAME	Customer's last name
EMAIL	Customer email address
PHONE_NUMBER	Customer contact number
PASSWORD	Customer account password
GENDER	Customer gender
DATE_OF_BIRTH	Customer date of birth
ADDRESS	Customer delivery address
CITY	Customer city
STATE	Customer state
PINCODE	Customer postal code
REGISTRATION_DATE	Date of account creation

PRODUCT TABLE:

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Brand	Product brand
Category_ID	Product category ID
Description	Product description
Price	Product price
Discount	Product discount
Stock_Quantity	Available stock
Size	Product size
Color	Product color
Image_URL	Product image
Rating	Customer rating

2. Define Primary and Foreign Key Relationships:

Category → Product

- CATEGORY.CATEGORY_ID → Primary Key
- PRODUCT.PRODUCT_ID → Primary Key
- PRODUCT.CATEGORY_ID → Foreign Key referencing CATEGORY.CATEGORY_ID
- Relationship: One-to-Many (1:M)

Meaning: One category can have many products, but each product belongs to one category.

1. Table creation:

CREATE TABLE CATEGORY

```
SQL> CREATE TABLE AjioCategory (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(50) NOT NULL,  
4     Category_Description VARCHAR2(100)  
5 );
```

Table created.

CREATE TABLE PRODUCT

```
SQL> CREATE TABLE AjioProduct (  
2     Product_ID NUMBER PRIMARY KEY,  
3     Product_Name VARCHAR2(100) NOT NULL,  
4     Category_ID NUMBER,  
5     Price NUMBER(10,2),  
6     Stock NUMBER,  
7     CONSTRAINT fk_ajio_category  
8     FOREIGN KEY (Category_ID)  
9     REFERENCES AjioCategory(Category_ID)  
10 );
```

Table created.

2. INSERTING VALUES INTO CATEGORY TABLE

```
SQL> INSERT INTO AjioCategory VALUES (1, 'Men Fashion', 'Men clothing and accessories');
1 row created.

SQL> INSERT INTO AjioCategory VALUES (2, 'Women Fashion', 'Women clothing and accessories');
1 row created.

SQL> INSERT INTO AjioCategory VALUES (3, 'Footwear', 'Shoes, sandals and footwear');
1 row created.

SQL> INSERT INTO AjioCategory VALUES (4, 'Accessories', 'Fashion accessories');
1 row created.

SQL> INSERT INTO AjioCategory VALUES (5, 'Kids Fashion', 'Clothing and accessories for kids');
1 row created.
```

3. INSERTING VALUES INTO PRODUCT TABLE

```
SQL> INSERT INTO AjioProduct VALUES (101, 'Men Casual Shirt', 1, 999.00, 50);
1 row created.

SQL> INSERT INTO AjioProduct VALUES (102, 'Women Kurti', 2, 799.00, 40);
1 row created.

SQL> INSERT INTO AjioProduct VALUES (103, 'Running Shoes', 3, 1499.00, 30);
1 row created.

SQL> INSERT INTO AjioProduct VALUES (104, 'Leather Wallet', 4, 599.00, 25);
1 row created.

SQL> INSERT INTO AjioProduct VALUES (105, 'Kids T-Shirt', 5, 499.00, 35);
1 row created.
```

4. DISPLAY CATEGORY RECORDS

```
CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
      1 Men Fashion
Men clothing and accessories

      2 Women Fashion
Women clothing and accessories

      3 Footwear
Shoes, sandals and footwear

CATEGORY_ID CATEGORY_NAME
-----
CATEGORY_DESCRIPTION
-----
      4 Accessories
Fashion accessories

      5 Kids Fashion
Clothing and accessories for kids
```

5. DISPLAY PRODUCT RECORDS

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID PRICE STOCK
-----
| | | 101
Men Casual Shirt
| | | 1 999 50
| | | 102
Women Kurti
| | | 2 799 40

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID PRICE STOCK
-----
| | | 103
Running Shoes
| | | 3 1499 30
| | | 104
Leather Wallet
```

6. Data deletion:

Delete product records

```
SQL> DELETE FROM AjioProduct  
| 2 WHERE Product_ID = 105;
```

```
1 row deleted.
```

```
SQL> DELETE FROM AjioProduct  
| 2 WHERE Product_ID = 104;
```

```
1 row deleted.
```

7. Data Updation:

Update product price and stock

```
SQL> UPDATE AjioProduct  
| 2 SET Price = 1099  
| 3 WHERE Product_ID = 101;
```

```
1 row updated.
```

```
SQL> UPDATE AjioProduct  
| 2 SET Stock = 60  
| 3 WHERE Product_ID = 103;
```

```
1 row updated.
```

8. CATEGORY WISE PRODUCT DETAILS

```
SQL> SELECT
  2     c.Category_Name,
  3     p.Product_ID,
  4     p.Product_Name,
  5     p.Price,
  6     p.Stock
  7 FROM AjioCategory c
  8 JOIN AjioProduct p
  9 ON c.Category_ID = p.Category_ID
 10 ORDER BY c.Category_Name;
```

9. CATEGORY WISE PRODUCT COUNT

```
SQL> SELECT
  2     c.Category_Name,
  3     COUNT(p.Product_ID) AS Total_Products
  4 FROM AjioCategory c
  5 LEFT JOIN AjioProduct p
  6 ON c.Category_ID = p.Category_ID
  7 GROUP BY c.Category_Name
  8 ORDER BY c.Category_Name;
```

CATEGORY_NAME	TOTAL_PRODUCTS
Accessories	0
Footwear	1
Kids Fashion	0
Men Fashion	1
Women Fashion	1